Modeling the Effect of Pitch Sequencing and Batter-Specific Adjustments on Swing-and-Miss Outcomes in MLB

Project Report 1

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1 Introduction

The introduction of the Statcast tracking system in Major League Baseball (MLB) in 2015 fundamentally changed how teams evaluate pitching performance. Every pitch thrown in an MLB game is now measured across dozens of metrics—release speed, spin rate, movement profile, release point, and extension—captured at the millisecond level (Major League Baseball, 2024). This wealth of data has fueled a shift in pitcher development strategy, with front offices increasingly investing in data-driven approaches to maximize the effectiveness of each pitch in a pitcher's arsenal.

At the core of modern pitching analytics lies a central question: *what attributes make a pitch difficult to hit?* Intuitively, pitches with higher velocity, greater movement, and longer extension should be harder to contact. However, standard logistic regression models that predict swing-and-miss (whiff) probability based solely on these physical characteristics often produce inconclusive results. In such models, the probability of a whiff, $p_i = P(Y_i = 1)$, is typically modeled as:

$$\log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1\,\texttt{Speed}_i + \beta_2\,\texttt{IVB}_i + \beta_3\,\texttt{HB}_i + \beta_4\,\texttt{Ext}_i + \beta_5\,\texttt{Arm}_i \tag{1}$$

where `IVB` denotes induced vertical break, `HB` horizontal break, `Ext` release extension, and `Arm` arm angle. Previous studies using this framework have found that main effects and simple interactions are often statistically insignificant or fail to explain the observed variation in outcomes.

The fundamental limitation of this approach is that it treats each pitch as an independent, context-free event, ignoring two critical sources of variation: **sequencing** and **batter heterogeneity**. First, pitches are part of a strategic sequence within a plate appearance. A pitch's effectiveness depends not only on its own physical properties but also on the buildup leading to it. For example, a slider may induce more whiffs when preceded by a fastball, because the speed differential disrupts the batter's timing—a phenomenon known as pitch tunneling (Toporek, 2017). Second, batters differ substantially in their baseline contact rates. Elite contact hitters rarely swing

and miss regardless of pitch quality, while high-strikeout batters may whiff at average pitches. Ignoring these batter-specific tendencies introduces unmodeled heterogeneity that weakens inference about the true relationship between pitch quality and outcomes.

To address these limitations, we propose a framework that incorporates both pitch sequencing and batter heterogeneity. Specifically, our research question is:

How do pitch sequencing (the type, location, and result of the previous pitch) and batter-specific tendencies jointly influence the probability of a swing-and-miss, beyond the physical characteristics of the pitch itself?

We pursue three objectives:

1. **Pitch Sequence Effects:** Investigate whether the previous pitch type and result significantly affect the whiff probability of the current pitch, and identify which sequence combinations (e.g., Fastball → Slider) show the largest effects.

2. **Batter Skill vs. Pitch Quality:** Implement a Generalized Linear Mixed Model (GLMM) with batter random effects to separate variance attributable to the pitch from variance attributable to the batter's contact ability (Gelman and Hill, 2006; Bolker et al., 2009).

3. **Model Comparison:** Evaluate whether models incorporating sequencing and mixed effects provide better fit and predictive power compared to the baseline physical-characteristics model, using likelihood ratio tests and BIC (McCullagh and Nelder, 2019).

In this first report, we describe the dataset, outline our data cleaning procedures, and present the results of an exploratory data analysis that motivates the modeling choices above.

2 Data Description

2.1 Data Source and Collection

Our dataset consists of pitch-level observations from the 2025 MLB regular season, obtained from the Statcast tracking system via the `pybaseball` Python module (Peta and Contributors, 2024). Statcast uses a combination of radar and optical tracking to record high-resolution measurements for every pitch thrown in MLB games. We retrieved approximately 711,897 pitch records spanning the full 2025 regular season (March–September).

2.2 Unit of Observation and Response Variable

The unit of observation is an individual pitch. For each pitch i, we define a binary response variable:

$$Y_i = \begin{cases} 1 & \text{if the batter swings and misses (whiff),} \\ 0 & \text{otherwise.} \end{cases}$$

A whiff is identified using the Statcast `description` field; specifically, we classify a pitch as a whiff when the description is `swinging_strike`, `swinging_strike_blocked`, or `foul_tip`.

2.3 Predictor Variables

We organize predictors into three categories:

(1) Physical Pitch Characteristics. These are the measurable properties of each individual pitch:

- `release_speed`: Release velocity (mph)
- `pfx_z`: Induced vertical break (inches)
- `pfx_x`: Horizontal break (inches)
- `release_extension`: Release extension toward home plate (feet)

(2) Sequencing Variables. To capture the strategic context of each pitch, we construct lagged variables within each plate appearance:

- Previous pitch type (e.g., fastball, slider, changeup)
- Previous pitch zone (Statcast zones 1–14)
- Previous pitch outcome (ball, called strike, swinging strike, foul, in-play)
- Interaction between the current and previous pitch types

Lag variables are constructed strictly within the same plate appearance. For the first pitch of an at-bat, previous-pitch fields are coded as a distinct "None" category to preserve the observation rather than discard it.

(3) Batter-Specific Effects. To account for heterogeneity in contact ability, batter identity will be incorporated as a random effect in the GLMM stage of modeling.

2.4 Data Cleaning

We apply the following filters to ensure data quality:

1. Remove non-competitive events (pitchouts, intentional balls).
2. Exclude pitches with missing physical measurements (`release_speed`, `pfx_x`, `pfx_z`, or `release_extension`).
3. Remove pitches with unrealistic release speeds (below 65 mph or above 105 mph), which indicate tracking errors.
4. Restrict to the nine most common pitch types: four-seam fastball (FF), sinker (SI), cutter (FC), slider (SL), sweeper (ST), curveball (CU), knuckle-curve (KC), changeup (CH), and splitter (FS).

After cleaning, our working dataset contains 702,384 pitches with an overall whiff rate of 12.0%.

3 Exploratory Data Analysis

Before proceeding to formal modeling, we conduct an exploratory analysis to characterize the key patterns in the data and motivate our modeling choices. All analysis and visualization is performed

in Python using the `matplotlib` (Hunter, 2007) and `seaborn` (Waskom, 2021) libraries.

3.1 Summary Statistics

Table 1 presents descriptive statistics for the four primary physical pitch characteristics. Release speed has a mean of approximately 90 mph with a standard deviation of 5.8 mph, reflecting the mix of fastballs (mid-to-upper 90s) and off-speed pitches (low-to-mid 80s) in a typical MLB game. Induced vertical break averages 6.9 inches but spans a wide range (-24.0 to 27.5 in.), capturing the contrast between pitches with heavy topspin (negative IVB, such as curveballs) and those with strong backspin (positive IVB, such as four-seam fastballs). Horizontal break also exhibits substantial variability (SD = 10.8 in.), reflecting the diversity of arm angles and pitch types in the dataset.

Variable	Mean	SD	Min	Max
Release Speed (mph)	89.5	5.8	65.2	104.2
Induced Vertical Break (in.)	6.9	8.6	-24.0	27.5
Horizontal Break (in.)	-1.6	10.8	-29.4	30.5
Release Extension (ft)	6.4	0.4	3.9	9.6

Table 1: Summary statistics for key physical pitch characteristics in the cleaned 2025 dataset.

3.2 Whiff Rates by Pitch Type

Not all pitch types are equally effective at generating swing-and-miss outcomes. Figure 1 displays the whiff rate for each pitch type in the dataset. Splitters lead with the highest whiff rate (17.8%), followed closely by knuckle-curves (16.4%) and sliders (16.2%). Off-speed and breaking pitches consistently outperform fastball variants: the three highest-whiff-rate pitch types are all non-fastball pitches, while sinkers produce the lowest whiff rate at just 6.3%. This pattern is expected, as off-speed and breaking pitches are designed to deceive the batter through movement and speed differential rather than pure velocity. Table 2 provides the corresponding counts.

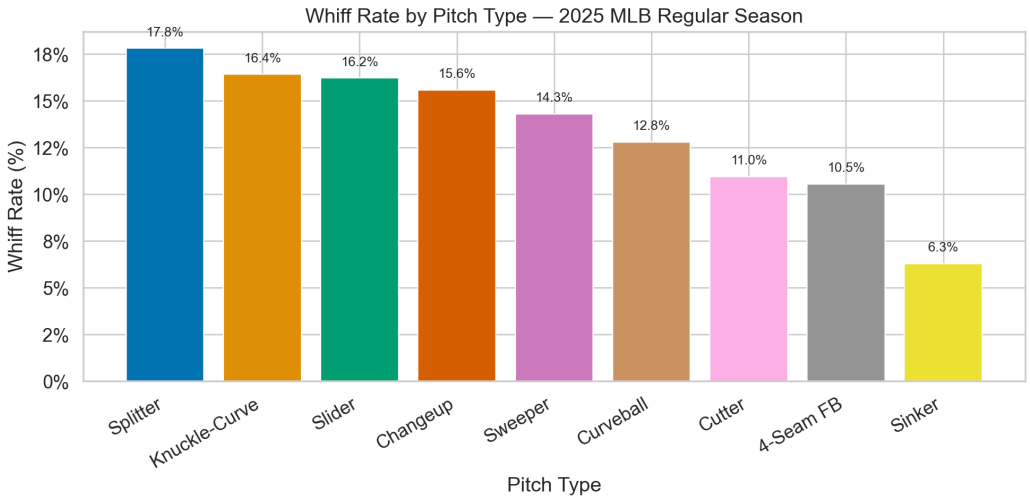

Figure 1: Whiff rate by pitch type in the 2025 MLB regular season. Breaking balls and off-speed pitches consistently show higher whiff rates than fastball variants.

Pitch Type	Total Pitches	Whiffs	Whiff Rate (%)
Splitter	23,131	4,119	17.8
Knuckle-Curve	12,081	1,983	16.4
Slider	104,622	16,964	16.2
Changeup	72,963	11,368	15.6
Sweeper	50,918	7,284	14.3
Curveball	48,248	6,176	12.8
Cutter	53,896	5,903	11.0
4-Seam FB	225,957	23,830	10.5
Sinker	110,568	6,952	6.3

Table 2: Whiff rates by pitch type in the 2025 MLB regular season.

3.3 Physical Pitch Characteristic Distributions

Figure 2 compares the distributions of release speed, induced vertical break, horizontal break, and release extension between whiff and non-whiff outcomes. Several patterns emerge. Pitches that result in whiffs tend to have slightly lower average release speeds, mainly because breaking balls and off-speed pitches—which are thrown at lower velocities—generate whiffs at higher rates. The induced vertical break distribution for whiffs is shifted toward lower (and more negative) values compared to non-whiffs, consistent with the prevalence of sliders, sweepers, and curveballs—pitches with downward break—among swing-and-miss outcomes. The horizontal break distribution for whiffs shows heavier tails, reflecting the extreme lateral movement of sliders and sweepers. Release extension distributions are largely overlapping between the two groups. Overall, these distributional

differences, while visible, are not dramatic for any single variable in isolation, reinforcing the need for multivariate modeling.

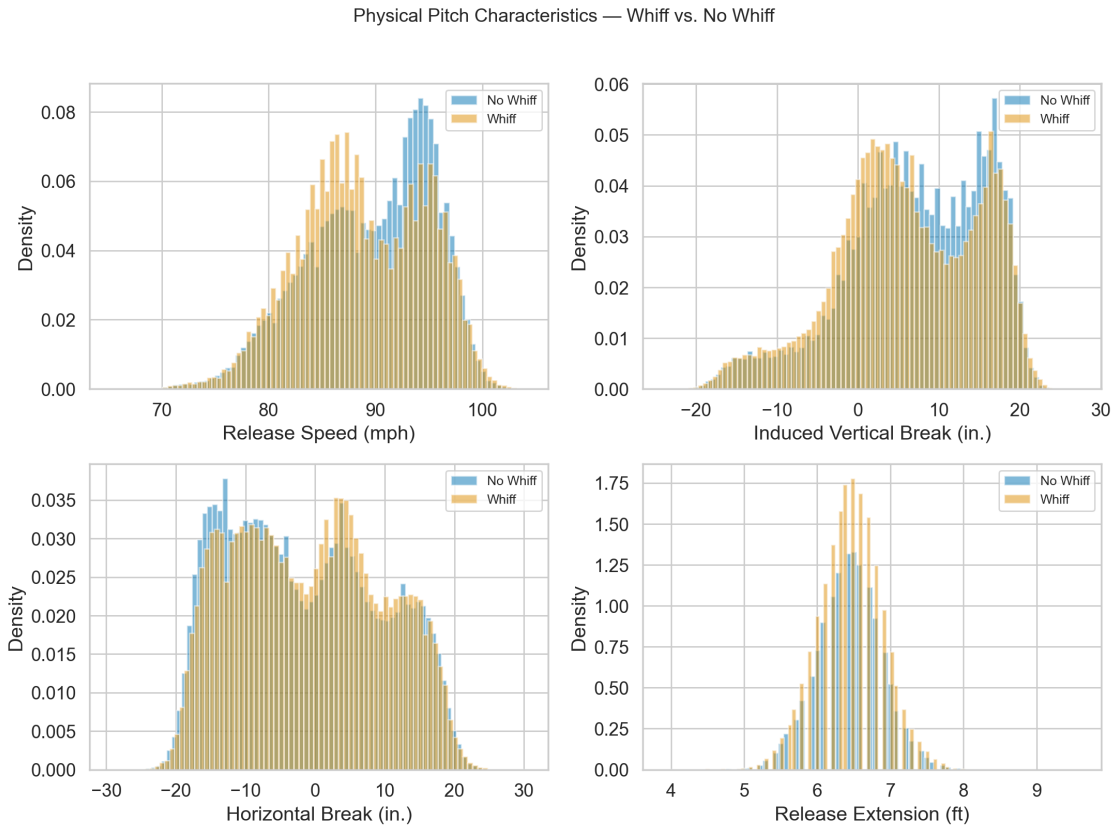

Figure 2: Density plots of physical pitch characteristics, stratified by outcome (whiff vs. no whiff). Differences across individual variables are subtle, motivating multivariate approaches.

3.4 Pitch Sequencing Patterns

A central hypothesis of this study is that pitch effectiveness depends on the preceding pitch. Figure 3 presents a heatmap of whiff rates organized by previous pitch type (rows) and current pitch type (columns). Several noteworthy patterns are visible. Splitters and knuckle-curves thrown after curveballs achieve the highest whiff rates in the heatmap (20.1% and 19.9%, respectively), suggesting that batters struggle to adjust to a second consecutive breaking ball with different movement profiles. Sequences from sinkers or four-seam fastballs to splitters also produce elevated whiff rates (19.2% and 16.0%), consistent with the pitch tunneling hypothesis: when consecutive pitches appear similar out of the pitcher's hand but diverge in trajectory near the plate, the batter's ability to adjust is compromised. Conversely, sinkers preceded by any pitch type consistently show the lowest column-wise whiff rates (5.7%–8.5%), suggesting that batters find sinkers easier to put in play regardless of sequence. Fastball variants (four-seam fastballs, cutters) thrown after breaking balls tend to have relatively moderate whiff rates (10–13%), suggesting that batters may sit on velocity after seeing off-speed pitches. These findings strongly motivate the inclusion of sequencing

variables and pitch-type interaction terms in the logistic regression model specified in Equation (1).

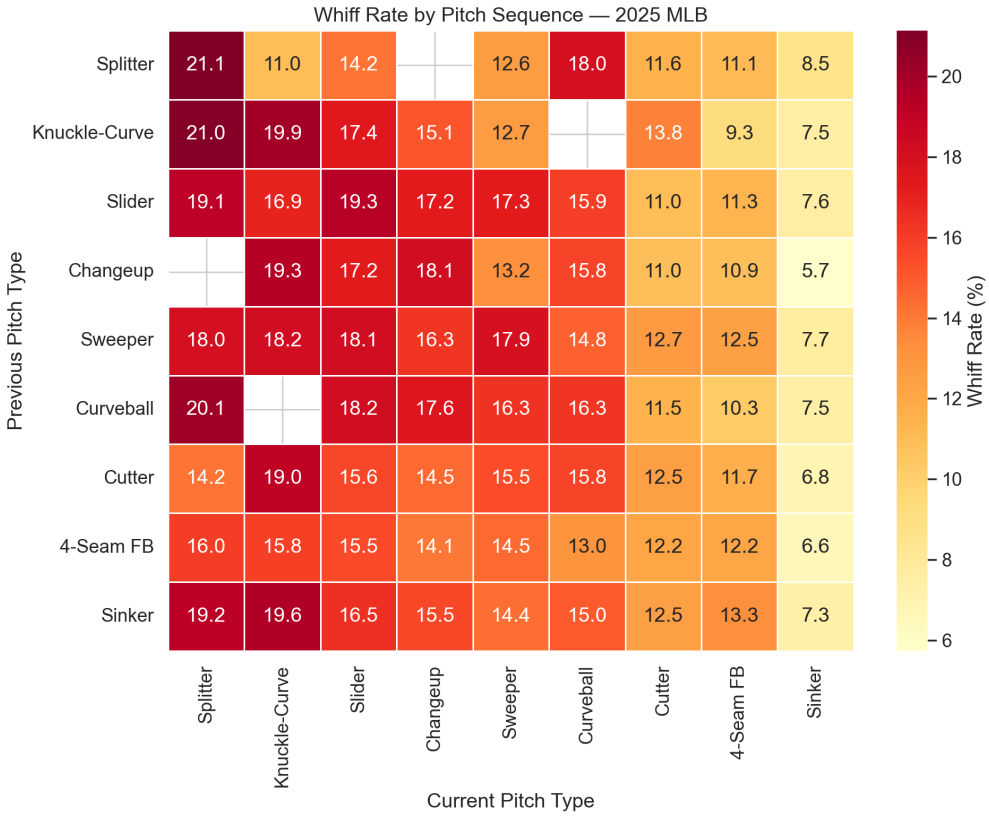

Figure 3: Heatmap of whiff rates by pitch sequence. Rows represent the previous pitch type; columns represent the current pitch type. Cells with fewer than 100 observations are excluded. Fastball-to-breaking-ball sequences show elevated whiff rates.

3.5 Batter Heterogeneity

To examine whether batter identity is an important source of variation, we compute per-batter whiff rates for all batters who faced at least 200 pitches during the season (531 batters). Figure 4 shows the distribution of these batter-level whiff rates. The distribution is approximately normal with a pronounced spread: batter-level whiff rates range from roughly 2.4% to 23.2%, illustrating the large degree of heterogeneity in contact ability across MLB hitters. This variability confirms that batter identity is an important grouping factor and justifies the use of batter-level random intercepts in the planned GLMM.

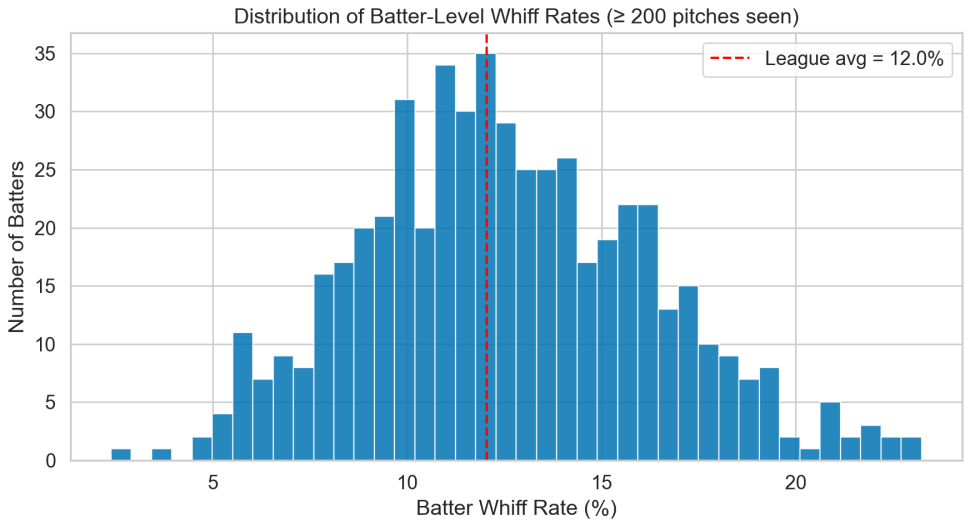

Figure 4: Distribution of batter-level whiff rates (batters with ≥ 200 pitches seen). The dashed red line marks the overall league whiff rate. Substantial heterogeneity motivates the use of batter random effects.

3.6 Predictor Correlations

Figure 5 presents the pairwise correlations among the continuous predictors and the binary whiff outcome. The most notable inter-predictor relationship is between release speed and induced vertical break ($r = 0.72$), which is expected because high-velocity fastballs generate strong backspin and thus higher vertical break. This substantial correlation warrants monitoring via Variance Inflation Factors (VIFs) in the modeling stage to guard against multicollinearity. Release speed and horizontal break are negatively correlated ($r = -0.30$), reflecting the tendency for slower breaking pitches to exhibit greater lateral movement. Critically, none of the individual correlations with the whiff outcome exceed $|r| = 0.05$ in magnitude—release speed ($r = -0.05$), IVB ($r = -0.04$), HB ($r = 0.02$), and extension ($r = 0.01$)—confirming that no single physical characteristic is a dominant predictor of swing-and-miss outcomes. This reinforces the need for richer models that incorporate sequencing and batter heterogeneity.

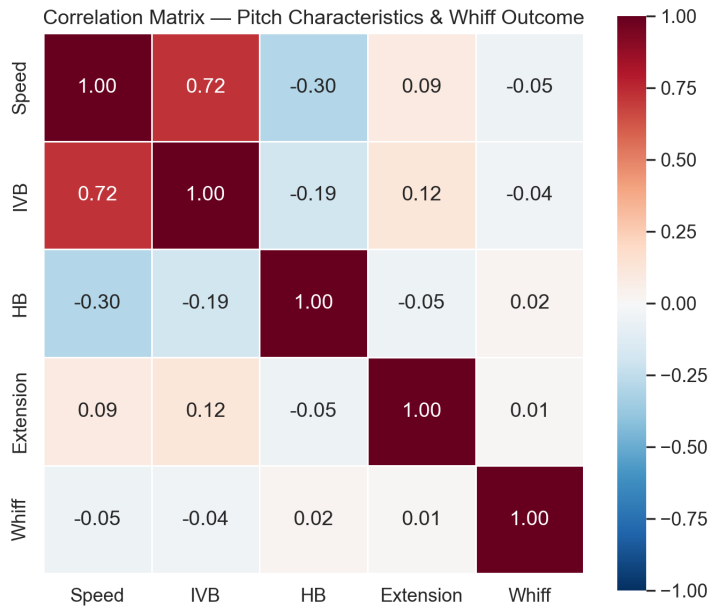

Figure 5: Correlation matrix of continuous pitch characteristics and the binary whiff outcome. Correlations with whiff are small in magnitude, supporting the need for richer models.

4 Discussion and Next Steps

The exploratory analysis reveals several findings that inform our modeling strategy. First, whiff rates vary substantially across pitch types, ranging from 6.3% (sinkers) to 17.8% (splitters), with off-speed and breaking pitches consistently outperforming fastball variants. Second, pitch sequencing appears to matter: certain transitions—such as curveball → splitter (20.1%) and sinker → knuckle-curve (19.6%)—produce whiff rates well above their individual pitch-type averages, consistent with the pitch tunneling hypothesis. Third, batter-level whiff rates span a wide range (2.4%–23.2% among batters with ≥ 200 pitches faced), confirming that batter identity is an important source of variation that should be accounted for in modeling. Fourth, no single physical pitch characteristic is strongly correlated with whiff outcomes (all $|r| \leq 0.05$), underscoring the need for multivariate models that incorporate both pitch-level and context-level information.

In the next phase of this project, we will proceed with the three-model framework outlined in the introduction:

1. A **baseline logistic regression** using physical pitch characteristics only (Equation 1),
2. A **sequencing model** that adds lagged pitch variables and pitch-type interactions, and
3. A **GLMM** that incorporates batter random intercepts.

We will compare these models using likelihood ratio tests, BIC, and out-of-sample predictive accuracy to determine whether sequencing and batter heterogeneity provide meaningful improvements over the baseline.

References

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